



CREW CHANGE RISK COPILOT

AI-Supported Decision Support for Smarter Crew Change Operations

From Validated PoC to MVP Architecture & Pilot Strategy

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CREW CHANGE OPERATIONS ARE COMPLEX — BUT ATTENTION IS LIMITED

Vessel Schedule

Operational timing

Crew Readiness

Availability and readiness

Travel & Connections

Flights and connections

External Disruptions

Weather and airport context

MULTIPLE INTERCONNECTED SIGNALS

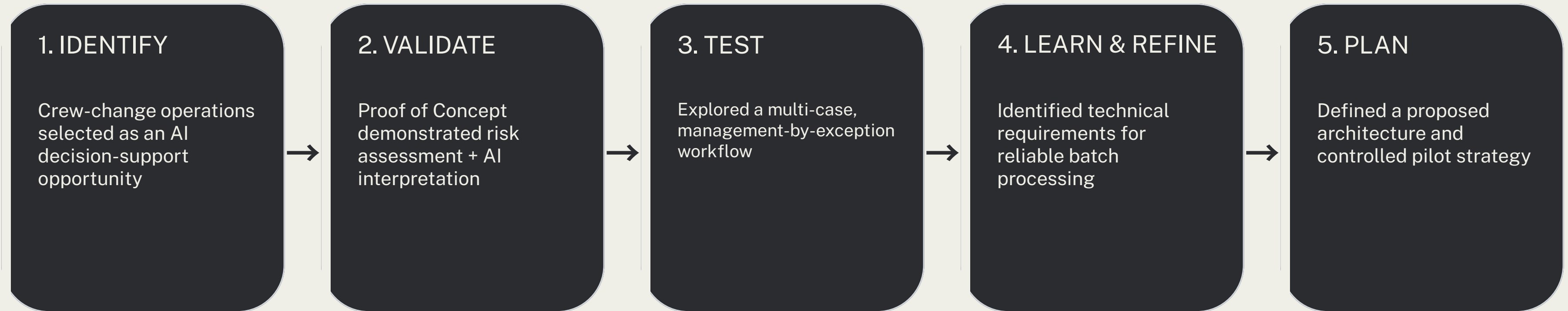


WHICH CASES REQUIRE ATTENTION FIRST?

Crew Managers must review multiple upcoming crew-change cases and determine which ones require attention first — often by combining information from different sources.

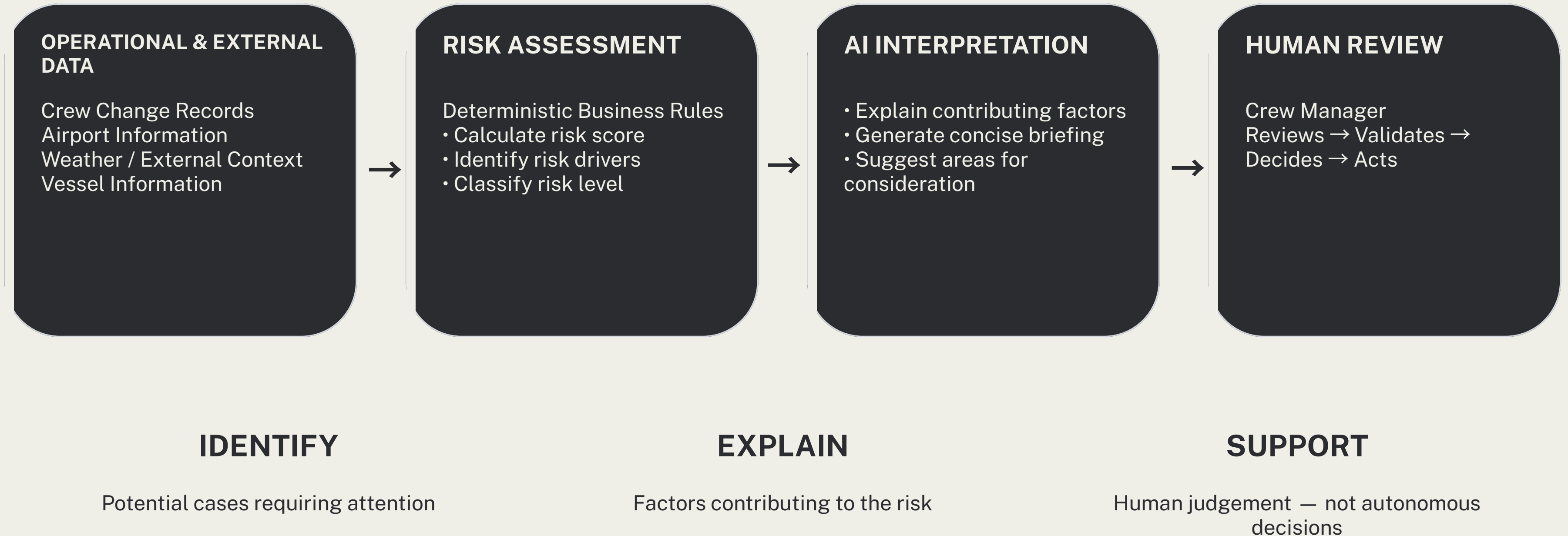
The challenge is not simply a lack of information — it is knowing where human attention is needed.

FROM INITIAL CONCEPT TO A CLEAR IMPLEMENTATION PATH

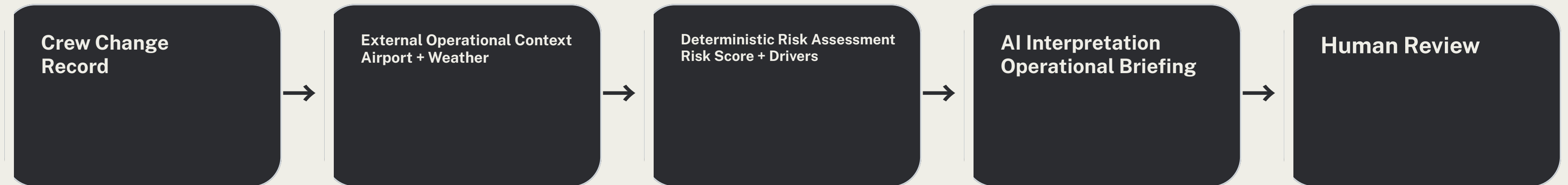


The business problem remained consistent. The technical approach evolved based on practical implementation findings.

CREW CHANGE RISK COPILOT: AI-SUPPORTED MANAGEMENT BY EXCEPTION



PROOF OF CONCEPT: VALIDATING THE CORE WORKFLOW



✓ Data Integration

Combined crew-change information with external operational context.

✓ Risk Assessment

Applied deterministic rules to calculate risk levels and identify risk drivers.

✓ AI Interpretation

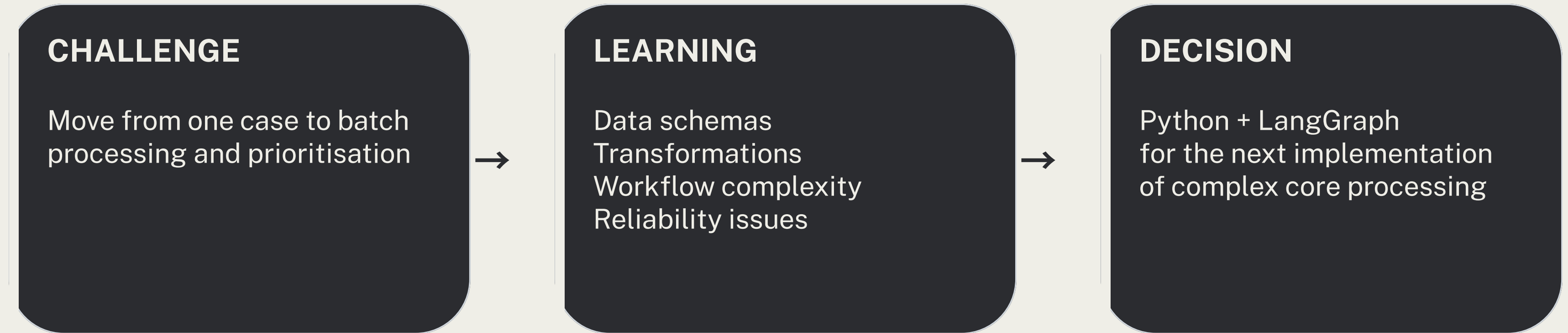
Generated a concise AI-assisted operational briefing based on structured risk information.

✓ Human-in-the-Loop

Kept the final operational decision with the Crew Manager.

The PoC validated the core concept: structured operational signals can be transformed into an explainable AI-assisted briefing for human review.

SCALING THE WORKFLOW: A KEY TECHNICAL FINDING



The business use case remained valid. The implementation approach evolved based on practical technical findings.

Key finding: do not continue adding complexity to the low-code workflow when stronger schema control, validation and batch processing are required.

BUSINESS CASE: CONSERVATIVE ROI

Only avoided disrupted crew-change costs are monetised at this stage.

€35,000

Initial investment
Planning assumption

€4,000

Annual operating cost
Estimated

€18,750

Direct annual benefit
15 avoided cases × €1,250

ROI OUTLOOK

12 MONTHS

Total cost: €39,000
Direct benefit: €18,750
-51.9% ROI

36 MONTHS

Total cost: €47,000
Direct benefit: €56,250
+19.7% ROI

Break-even: ≈32 avoided problematic crew-change cases to recover first-year costs through direct cost avoidance alone.

COMPLIANCE BY DESIGN

Human Control • Transparency • Data Protection

EU AI ACT — CURRENT ASSESSMENT: NON-HIGH-RISK

- Supports operational prioritisation
- Does not make employment decisions
- Does not autonomously decide whether someone may travel or work
- Final decision remains with an authorised human

Focus: Transparency + proportionate governance

GDPR — CURRENT MVP VS PRODUCTION

Current MVP: Synthetic / public data

Production: Potentially identifiable crew, travel and operational information

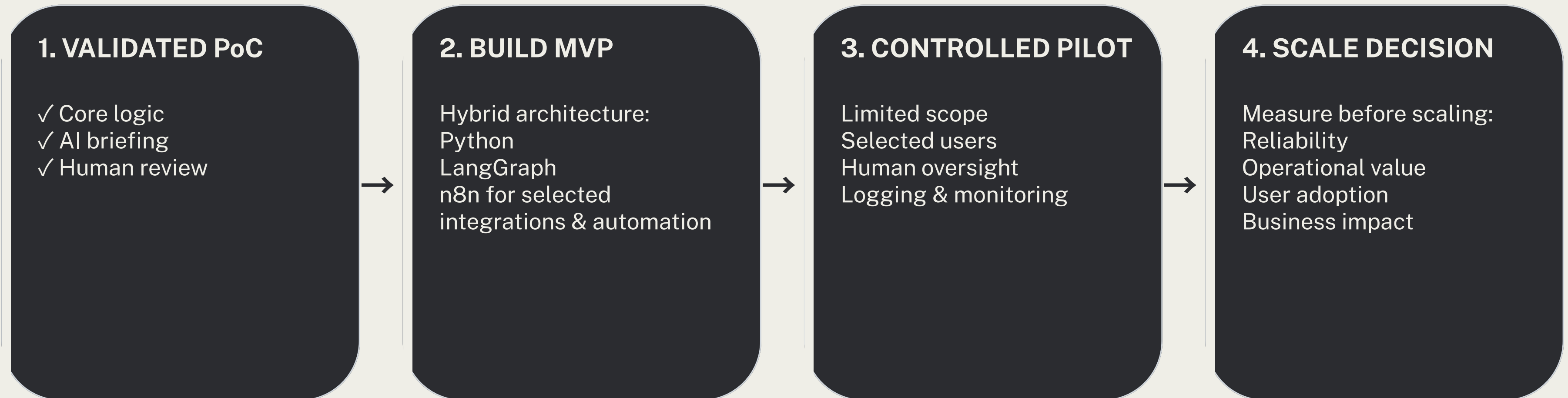
Controls: Data minimisation • Pseudonymisation where possible • Role-based access • Retention & deletion controls • Vendor / transfer assessment

AI ANALYSES → AI FLAGS & EXPLAINS → HUMAN REVIEWS → HUMAN DECIDES

Compliance is built into the operating model — not added after the technology is built.

STRATEGIC IMPLEMENTATION ROADMAP

A phased path from validated concept to operational deployment



Hybrid principle: Move complex core processing into code while retaining low-code automation where it provides practical value.

Progress through evidence-based validation — not technology-driven scaling.

CREW CHANGE RISK COPILOT

From Concept to a Clear Path Forward

✓ CONCEPT VALIDATED

Proof of Concept demonstrated the core workflow and AI-assisted briefing.

✓ BUSINESS CASE PROMISING

Conservative model becomes positive over three years; additional benefits require pilot validation.

✓ IMPLEMENTATION PATH DEFINED

Hybrid MVP architecture followed by a controlled pilot before wider deployment.

RECOMMENDATION

PROCEED WITH A FOCUSED MVP IMPLEMENTATION AND CONTROLLED PILOT.

Validate: Technical Reliability • Operational Value • User Adoption • Business Impact

AI supports human judgement — it does not replace it.